Walter Karshat

Software Engineer and Executive | AI Coding Agents and Tooling

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Portfolio · GitHub · LinkedIn

Systems software engineer, company founder, and CTO / VP level engineering executive at startups and midsize technology companies. Repeatedly recruited and built engineering teams while staying hands-on in product architecture and code development.

Current work focuses on building tools for visibility into how large language model (LLM) coding agents and harnesses operate internally. Hands-on experience and strong background in deploying complex blockchain configurations, designing Android devices to demanding carrier specifications, and authoring optimized embedded software, infrastructure telecom and high-speed networking equipment, and real-time systems for clinical applications.

Recent AI and developer tooling

Codess — AI coding harness observability | 2026

- Developed Python tooling to discover, decode, and query local session records from Claude Code, Codex, and Cursor. Codess
- Mapped prompts, model turns, tool calls, Model Context Protocol (MCP) interactions, and permissions into a common schema while retaining links to original vendor records. CoSchema
- Added diagnostics to expose gaps in session records. Structured exports support cross-harness comparison, failure analysis, and instruction-following evaluation.

Additional projects | 2026

- **Fear AI** — Built a basis for a publication site to cover in-depth factual practicalities of AI tooling, employment trends, societal trust, and requisite infrastructure buildout. Fear AI
- **uniblake** — Developed portable BLAKE2b implementations for efficient shared-prefix hashing, with caller-controlled scheduling and conformance verified against an independent reference. Uniblake

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Selected experience

- **KIN Systems | Principal | 1996–present** — Instituted processes and personally led product specification, architecture design and full SDLC development and validation across cloud, mobile, IoT, and blockchain environments.
- **Alpha Block | CTO | 2017–2020** — Blockchain investment intelligence. Served as a POA Network validator and SPIRIT Blockchain Technologies advisor.
- **Motorola | Distinguished Member of Technical Staff | 2006–2014**
  - Designed and coded a stable real-time audio path between Qualcomm's real-time OS and Android to support Wi-Fi calling. Modeled network delay and jitter, measured packet-loss concealment effect on call quality.
  - Honed Wi-Fi performance and power management. Coordinated Wi-Fi Alliance certification testing and Bluetooth coexistence compliance.
  - Led the transition to Android development and build tooling for T-Mobile, AT&T, and LATAM handsets.
- **Prior Engineering and Leadership** — Designed Linux traffic management and voice channel allocation for satellite broadband at Tachyon Networks. Engineered voice transcoding and security ASIC system software and drivers at GlobespanVirata and at Hifn. Led CT/MRI workstation development at Pixar. PICS 2000

Technical skills and education

Python, C, C++, JavaScript, TypeScript, Rust, Linux, Android

Schema design and translation, MCP protocol compliance validation

UC Berkeley — BA, Computer Science and Physics, with honors